Bulletin of Network Engineer and Informatics
E-ISSN 2986-8017 | P-ISSN 2987-4858
https://bufnets.tech

Vol. 4 No 1 – July 2026
DOI: 10.59688/738832

SONG POPULARITY PREDICTION ON THE SPOTIFY PLATFORM USING MULTIPLE LINEAR REGRESSION WITH MIN-MAX NORMALIZATION

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Received : 28 May 2026	Accepted : 06 July 2026	Published : 10 July 2026

Abstract

This study critically investigates whether three commonly used audio-level features of a song, duration, tempo, and energy, are by themselves, statistically capable of predicting song popularity on the Spotify platform. Using a multiple linear regression model estimated via Ordinary Least Squares (OLS) on 849 songs retrieved from Kaggle 'Spotify Tracks Dataset', a publicly available collection spanning multiple genres and release periods, filtered to exclude duplicate records and songs with a popularity score of zero, we show that this is not the case. Prior to modeling, Min-Max normalization was applied to standardize variable scales. The model yields the equation $Y = 42.4073 + (-5.2462)\cdot X1 + (1.0306)\cdot X2 + (-4.1247)\cdot X3$, but the overall F-test is not significant ($F = 1.1789 < F\text{-table} = 2.6154$, $p = 0.3167$), and none of the three predictors reach individual significance (all $|t| < 1.53$, $p > 0.12$). Pairwise correlations between each predictor and popularity are negligible ($r = -0.037$ for duration, $r = -0.006$ for tempo, $r = -0.053$ for energy), confirming that the linear structure captures essentially no signal in this dataset. We conclude that duration, tempo, and energy have effectively zero standalone predictive capability for Spotify popularity, and that normalization, while statistically inert with respect to R^2 and RMSE, remains necessary for producing comparable, interpretable coefficients. As a secondary, practical contribution, the analysis pipeline was deployed as an interactive web application (SpotiPredict, built with Python Flask) that exposes the full normalization and regression computation transparently to the user.

Keywords: *Multiple linear regression, falsification study, audio feature limitations, Min-Max normalization, song popularity, Spotify*

1. INTRODUCTION

The way people listen to music has fundamentally changed due to digital technology advancements, shifting from physical media to internet-based streaming platforms. Spotify, as one of the largest music streaming platforms in the world, functions not only as entertainment but also as an abundant source of numerical data. Each song possesses quantitatively measurable audio metadata, including duration, tempo, energy, and popularity score[1], [2].

Song popularity is a key indicator that influences label business decisions, artist revenue, and marketing strategies. However, objectively predicting song popularity is challenging, as it is affected by intrinsic factors (audio characteristics) and extrinsic factors (promotion and social media trends). Data-driven approaches are considered more effective because they produce more consistent and measurable estimations[3], [4], [5].

Multiple linear regression is well suited to testing, rather than merely fitting, the hypothesis that a defined set of variables explains a given outcome [6],[7]. Rather than attempting to maximize predictive accuracy through an extensive feature set, this study deliberately isolates three audio-intrinsic variables, duration, tempo, and energy, to test, in a controlled and falsifiable manner, whether audio metadata alone carries meaningful explanatory power over Spotify popularity. This design choice follows a falsification logic: if even a well-specified OLS model with properly normalized predictors fails to reach statistical significance, this constitutes evidence, not merely an absence of evidence, that intrinsic audio characteristics are insufficient predictors in the modern streaming ecosystem, and that extrinsic factors (artist identity, promotional activity, playlist placement, social trends) are necessary components of any viable popularity model [4],[15]. This reframes the contribution: the

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study does not propose a competing predictive model, but provides a quantitative, reproducible demonstration of the explanatory ceiling of audio-only linear models, motivating the field's shift toward socially and contextually enriched feature sets.

These studies collectively establish that comprehensive feature sets, particularly those incorporating social, contextual, or extended audio features such as valence and danceability, achieve materially higher predictive accuracy (e.g., up to 69% in [15]) than is achievable from a minimal intrinsic audio subset. The present study formally tests, rather than assumes, the magnitude of this gap by isolating duration, tempo, and energy as the sole predictors and evaluating whether their combined linear effect is statistically distinguishable from zero.

2. RESEARCH METHODOLOGY

2.1 Research Procedure

The research stages include: (1) data collection from Kaggle; (2) preprocessing: data cleaning and outlier removal; (3) Min-Max normalization of independent variables; (4) construction of the multiple linear regression model using OLS via Python's Scikit-learn library; (5) classical assumption tests (Kolmogorov-Smirnov normality test, VIF multicollinearity test, and Glejser heteroscedasticity test); (6) model evaluation (R^2, Adjusted R^2, RMSE, F-test, t-test); and (7) model deployment in a Python Flask web application.

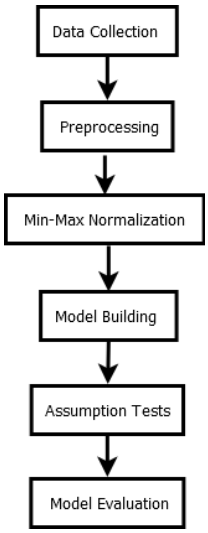


Figure 1. Research Flowchart

2.2 Dataset Collection

The dataset used is the "Spotify Tracks Dataset" from Kaggle in CSV format, consisting of 849 songs after preprocessing (removal of duplicate records and entries with popularity = 0). The independent variables are X_1 = Duration (ms), X_2 = Tempo (BPM), and X_3 = Energy (scale 0–1), while the dependent variable is Y = Popularity (scale 0–100). These three independent variables were selected because they represent numerically measurable audio characteristics that have been shown to correlate with popularity levels in previous studies[3],[9],[11],[12].

Table 1: Descriptive Statistics of Research Variables
[Source: Kaggle – Spotify Tracks Dataset, processed 2026]

Variable	Unit	Min	Max	Mean	Std.Dev
Duration (X_1)	ms	30,622.0	662,453.0	225,163.8	82,931.6
Tempo (X_2)	BPM	47.745	220.084	124.123	30.425
Energy (X_3)	0–1	0.0001	0.9960	0.6343	0.2593
Popularity (Y)	0–100	1.0	96.0	38.622	19.679

2.3 Preprocessing

The preprocessing stage involves two main steps. First, duplicate records are identified and removed to ensure each song is represented only once in the dataset. Second, records with a popularity score of 0 are excluded, as a value of zero typically indicates that the song has not been streamed or rated on the platform, making it uninformative for modeling purposes. After preprocessing, the dataset is reduced to 849 valid song records ready for further analysis[16].

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2.4 Min-Max Normalization

Prior to model construction, Min-Max normalization is applied to all three independent variables (Duration, Tempo, and Energy) to transform their values into a uniform range of [0, 1] using the formula[19], [20]:

$$X' = (X - X_min) / (X_max - X_min) \quad (1)$$

This step is necessary because the three variables differ substantially in scale: Duration is measured in milliseconds (30,622–662,453 ms), Tempo in beats per minute (47.745–220.084 BPM), and Energy in a decimal range (0–1). Without normalization, variables with larger numeric ranges would disproportionately dominate the regression coefficients, reducing their interpretability. Normalization ensures that each variable contributes comparably to the model[20].

2.5 Model Building

The multiple linear regression model is constructed using the Ordinary Least Squares (OLS) method implemented via Python's Scikit-learn library. Classical assumption tests, specifically the Kolmogorov-Smirnov normality test, the Variance Inflation Factor (VIF) multicollinearity test, and the Glejser heteroscedasticity test, were implemented using Python's SciPy library (scipy.stats) and Statsmodels library (statsmodels.stats.diagnostic), ensuring full reproducibility of the statistical validation pipeline. The normalized independent variables (X_1, X_2, X_3) are used as predictors and the popularity score (Y) as the dependent variable. OLS estimates the regression coefficients by minimizing the sum of squared residuals between predicted and actual popularity values, expressed as $\beta = (X^TX)^{-1} \cdot X^Ty$. The resulting equation takes the form[8]:

$$Y = a + b_1X_1 + b_2X_2 + b_3X_3 \quad (2)$$

where the intercept and coefficients are estimated from the 849 training data points.

2.6 Rationale for Linear Specification

Given that the Kolmogorov-Smirnov test rejected residual normality ($D = 0.05096$, $p = 0.02347$), a natural question is why a log-transformation or polynomial feature expansion was not applied. Two considerations motivated retaining the linear specification. First, the rejection of normality is most plausibly attributable to sample size rather than genuine non-normal structure: with $n = 849$, the Kolmogorov-Smirnov test is known to be highly sensitive, often rejecting H0 for residual deviations that are negligible in practical terms; the Central Limit Theorem supports the validity of coefficient inference at this sample size regardless. Second, and more fundamentally, the correlation matrix in Section 3.3 (Figure 3) shows that all three predictors exhibit near-zero linear correlation with popularity ($|r| \leq 0.053$). A logarithmic or polynomial transformation systematically reshapes the marginal distribution of a predictor, but it cannot manufacture an association where none exists in the joint distribution of (X, Y); transformations alter functional form, not statistical independence. Applying non-linear mappings here would not address the underlying absence of signal, and risks superficially improving in-sample fit while obscuring the substantive finding that audio-only features are uninformative for this outcome. We therefore retain the linear specification deliberately, as it provides the most transparent and falsifiable test of the no-effect hypothesis which is the explicit object of this study, per the reframed Introduction.

2.7 Model Evaluation

Model performance is evaluated using five metrics. (1) Coefficient of Determination (R^2), measures the proportion of variance in the popularity score explained by the model; a value closer to 1 indicates a better fit. (2) Adjusted R^2, a corrected version of R^2 that accounts for the number of predictors in the model, penalizing the addition of irrelevant variables. (3) Root Mean Square Error (RMSE), represents the average magnitude of prediction errors in the original popularity scale; lower values indicate higher accuracy. (4) F-test, determines whether the overall regression model is statistically significant by comparing the F-statistic against the F-table at a significance level of $\alpha = 0.05$. (5) t-test, evaluates the individual significance of each regression coefficient to determine whether each predictor variable has a significant effect on popularity [4],[15].

3. RESULT AND DISCUSSION

3.1 Effect of Min-Max Normalization

Min-Max normalization was applied to all three independent variables prior to model construction. The comparison shows that normalization does not alter the values of R^2, Adjusted R^2, or RMSE. However, it

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rescales the coefficients, making them more comparable across variables. Without normalization, coefficient b_1 (duration) is extremely small (−0.0000083) due to the large scale difference, rendering it difficult to interpret [7],[15].

Table 2: Model Comparison Before and After Normalization

Metric	Before	After	Remark
R^2	0.004168	0.004168	Unchanged
Adj. R^2	0.000632	0.000632	Unchanged
RMSE	19.6263	19.6263	Unchanged
Coef. b_1	−0.0000083	−5.2462	Scale changed
Coef. b_2	0.005980	1.0306	Scale changed
Coef. b_3	−4.1415	−4.1247	Scale changed

3.2 Multiple Linear Regression Model

The regression model constructed from 849 Spotify songs yields the following equation:

$$Y = 42{,}4073 - 5{,}2462 \cdot X_1 + 1{,}0306 \cdot X_2 - 4{,}1247 \cdot X_3 \quad (3)$$

The Coefficients in Equation (3) were derived through the Ordinary Least Squares (OLS) method, which solves the matrix equation $\beta = (X^TX)^{-1} \cdot X^Ty$. In the process, all 849 normalized song records were arranged into a matrix X of size 849×4, with the first column filled with ones to compute the intercept, while the actual popularity scores were arranged into a vectoryofsize849×1.The OLS algorithm simultaneously identified the four coefficient values that minimize the total sum of squared differences between predicted and actual popularity values across all 849 records, expressed as $SSE = \Sigma(Y_i - \hat{Y}_i)^2 = \text{minimum}$. It is important to note that these coefficient values are entirely determined by the content of the 849-song dataset, had the dataset contained different songs, the resulting coefficients would differ accordingly. Thecomputation was performed automatically using the LinearRegression module from Python's Scikit-learn library[6],[15]

Although Equation (3) reports point estimates for a, b_1, b_2, and b_3, the F-test in Section 3.4 indicates the model as a whole is not statistically distinguishable from a null (intercept-only) model ($F = 1.1789$, $p = 0.3167$), and the t-tests in Table 4 show that none of the three coefficients individually differ from zero at the 0.05 level (Duration: $t = -1.0183$, $p = 0.3088$; Tempo: $t = 0.2586$, $p = 0.7960$; Energy: $t = -1.5251$, $p = 0.1276$). Under these conditions, the signs and magnitudes of b_1, b_2, and b_3 are statistical noise rather than estimable effects, and we do not interpret them substantively. Instead, Sections 3.3–3.4 below examine why the model fails to detect any structure in this dataset.

3.3 Distributional and Correlation Analysis

We computed the skewness of the dependent variable and the pairwise Pearson correlations between each (normalized) predictor and popularity directly from the 849-record dataset:

Table 3: Distributional Skewness of Research Variables

Variable	Skewness	Interpretation
Popularity (Y)	0.046	Approximately symmetric NOT zero-inflated or heavily skewed
Duration (X1, norm.)	1.077	Moderately right-skewed (long-duration outliers)
Tempo (X2, norm.)	0.353	Approximately symmetric
Energy (X3, norm.)	−0.660	Moderately left-skewed (energy concentrated at higher values)

Contrary to a common assumption in popularity-prediction literature, the dependent variable in this dataset is not zero-inflated: only 5.9% of songs have popularity ≤ 5, and the distribution (Figure 2) is broadly multi-modal but centered near the sample mean (38.62), not collapsed near zero. This rules out zero-inflation as the explanation for the model's failure.

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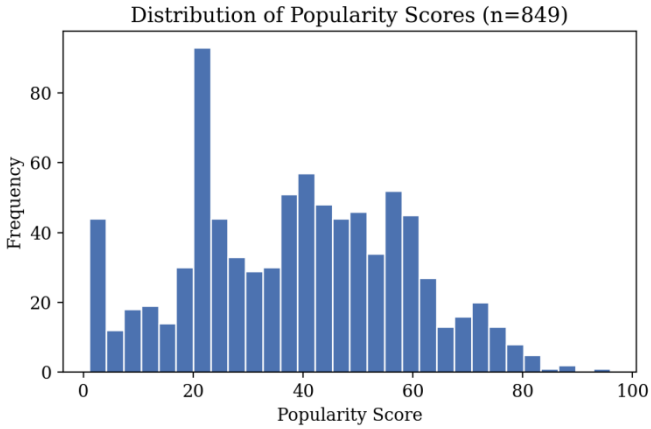


Figure 2. Distribution of Popularity Scores (n = 849)

The Pearson correlation matrix below makes the source of the failure explicit: every predictor–outcome correlation is close to zero.

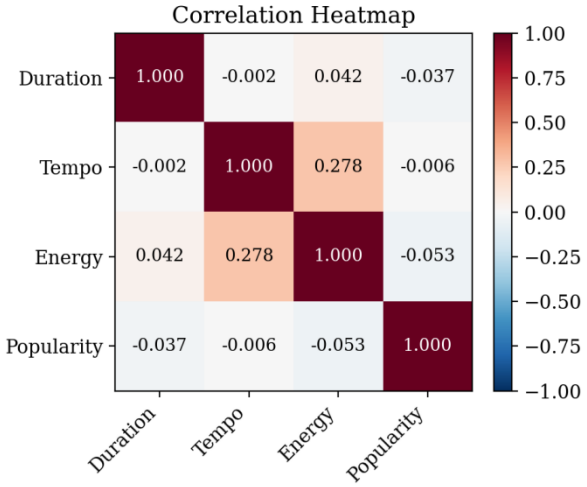


Figure 3. Correlation Heatmap of Normalized Predictors and Popularity

Pairwise scatter plots with fitted OLS lines (Figure 4) confirm this visually: in all three panels, the point cloud is essentially isotropic with respect to the x-axis, and the fitted regression line is nearly flat across the entire [0,1] normalized range. This is precisely the visual signature of an absent linear relationship the model is not merely weak, it is structurally blind to whatever process determines popularity in this dataset.

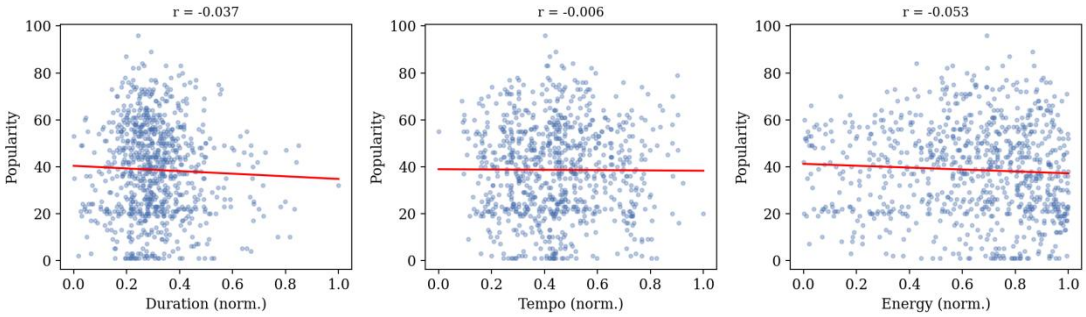


Figure 4. Scatter Plots of Normalized Predictors vs. Popularity, with OLS Fit Lines

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Together, these diagnostics show that the non-significant F-test and t-tests in Section 3.4 are not an artifact of model misspecification, distributional pathology, or an unlucky sample they are the direct, visually verifiable consequence of near-zero linear association between each predictor and the outcome in the underlying data.

3.4 Classical Assumption Tests

The Kolmogorov-Smirnov test shows that residuals are not normally distributed ($D=0.05096$, $p=0.02347 < 0.05$). However, based on the Central Limit Theorem, with $n=849$ the sampling distribution of coefficients approximates normality, making inference still valid. The VIF test reveals no multicollinearity (Duration: VIF=1.0020; Tempo: VIF=1.0843; Energy: VIF=1.0862; all < 10). The Glejser test shows no heteroscedasticity (all $p > 0.05$). Therefore, the assumptions of freedom from multicollinearity and homoscedasticity are satisfied[8],[16].

3.5 Model Evaluation

Table 4: Regression Model Evaluation Results

Metric	Value	Interpretation
R^2	0.004168	Model explains 0.42% of popularity variation
Adjusted R^2	0.000632	After correction for number of variables (k=3)
RMSE	19.6263	Average prediction error of ±19.63 points
F-statistic	1.1789	< F-table 2.6154; p=0.3167 (not significant)

The correlation and scatter-plot diagnostics in Section 3.3 corroborate the formal hypothesis tests: with $|r| \leq 0.053$ for every predictor, no linear model regardless of specification could be expected to achieve meaningful explanatory power from these three variables alone. This is consistent with Sebastien et al. [5], who report that audio features carry limited standalone predictive power, and contrasts with Saragih [15], who achieved 69% accuracy only after incorporating valence, danceability, and other engagement-adjacent features. The comparison is informative: the marginal predictive contribution of duration, tempo, and energy specifically isolated from the broader feature sets used in prior work appears to be negligible in this dataset, reinforcing that popularity on Spotify is governed predominantly by extrinsic factors rather than these three intrinsic audio descriptors.

3.6 Web Application Implementation

The SpotiPredict web application was developed using Python Flask to interactively implement the regression model. It consists of three main pages: (1) **Prediction Page** – users input duration, tempo, and energy values; the system performs Min-Max normalization and substitutes the values into the regression equation to produce a predicted popularity score along with step-by-step calculation details; (2) **Dataset Page** – displays all 849 songs with actual and predicted popularity scores, a search feature, and pagination; (3) **Model Information Page** – presents the complete regression equation, normalization parameters, coefficients, and evaluation results.

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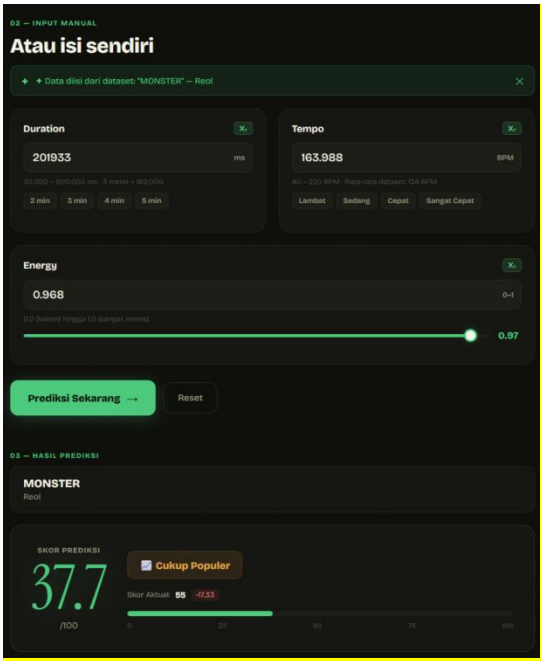


Figure 5. Prediction Page of SpotiPredict Web Application

Figure 5 shows the Prediction Page of the SpotiPredict application. Users are required to input three audio attribute values: Duration (in milliseconds), Tempo (in BPM), and Energy (on a scale of 0 to 1). Upon submission, the system automatically performs Min-Max normalization on the input values and substitutes them into the regression equation to compute a predicted popularity score. The page also displays a detailed step-by-step calculation, allowing users to trace the normalization and prediction process transparently.

Figure 6 presents the regression equation as shown within the application interface, confirming the model output accessible to end-users.

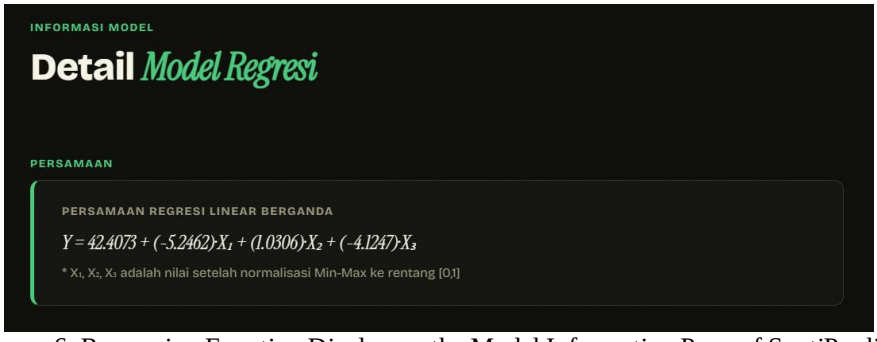


Figure 6. Regression Equation Display on the Model Information Page of SpotiPredict

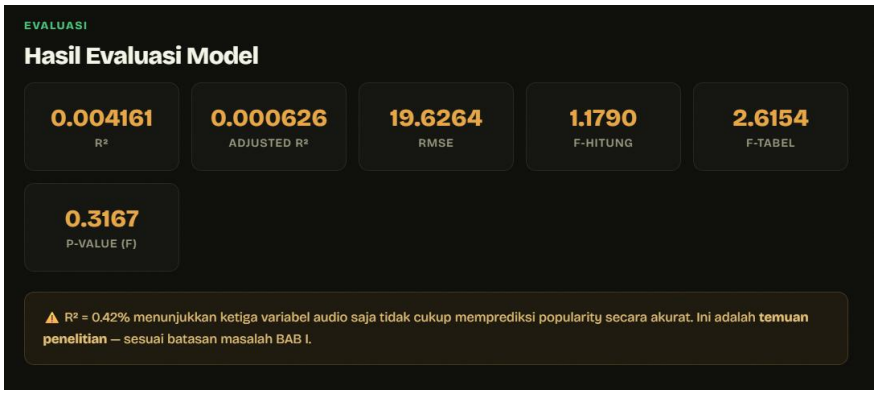


Figure 7. Model Information Page of SpotiPredict Web Application

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Figure 6 displays the regression equation as presented in the SpotiPredict application, showing the estimated coefficients derived from the OLS computation on 849 songs. Figure 7 displays the Model Information Page of SpotiPredict, which presents the complete regression equation, Min-Max normalization parameters, estimated coefficients (b_1, b_2, b_3), and full model evaluation results comprising R^2, Adjusted R^2, RMSE, F-statistic, and p-value. This page serves as a technical reference for users who wish to understand the underlying model structure and assess its statistical validity..

4. CONCLUSION

This study set out to test, rather than assume, whether three intrinsic audio variables duration, tempo, and energy carry meaningful explanatory power over Spotify song popularity, using Min-Max normalization and OLS-based multiple linear regression on 849 songs. The result is unambiguous: the overall model is not statistically significant ($F = 1.1789$, $p = 0.3167$), no individual coefficient differs from zero (all $p > 0.12$), and direct correlation analysis confirms negligible association between each predictor and popularity ($|r| \leq 0.053$). Distributional analysis further rules out zero-inflation or extreme skew in the outcome variable as confounding explanations. Taken together, these results constitute evidence that duration, tempo, and energy in isolation have effectively zero standalone predictive capability for Spotify popularity.

This finding should be read as a methodological caution rather than a null result to be dismissed: it demonstrates concretely that fitting an OLS model to a small set of intrinsic audio descriptors, without validating the presence of underlying linear structure first, risks producing coefficients that appear interpretable but are statistically meaningless. Researchers working with similarly minimal audio-only feature sets should report correlation/distributional diagnostics (as in Section 3.3) before interpreting regression coefficients substantively.

We additionally note that Min-Max normalization, while statistically inert with respect to R^2 and RMSE in this study, remains methodologically necessary: it is what allows the (statistically non-significant) coefficients to be placed on a comparable scale at all, and its absence would have made even the diagnostic exercise of this study harder to communicate.

Looking ahead, given the demonstrated absence of linear signal, we do not recommend simply expanding the variable set within an OLS framework, as this is unlikely to resolve the underlying issue if the audio–popularity relationship is non-linear or interaction-driven.Instead, future work should pursue a fundamental shift in modeling approach: (1) non-linear ensemble methods such as Random Forest or Gradient Boosting, which can capture threshold effects and interactions between audio features without requiring a pre-specified functional form; (2) deep learning architectures capable of learning latent representations directly from raw or higher-dimensional audio features; and (3) explicit incorporation of extrinsic, contextual variables artist popularity, genre, playlist placement, and release timing which prior work [15] suggests dominate the explanatory power that audio-only models like this one fail to achieve. As a secondary, practical output, the modeling pipeline developed here was deployed as SpotiPredict, an interactive Python Flask web application that transparently exposes the normalization and regression computation, providing a reusable foundation for testing future, more expressive modeling approaches on the same dataset.

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